

Nithil Eshwar Mani

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EDUCATION

- Master of Science (Data Science) - University at Buffalo 2025
- Bachelor of Engineering (Computer Science) - College of Engineering, Guindy 2023
- Certifications: AWS Certified Data Engineer Associate, AWS Certified ML Engineer Associate, Claude Certified Developer - Foundation

EXPERIENCE

Associate System Engineer, IBM - Bangalore, India May 2026 - Present

- Worked on developing and testing full stack applications using Python and Java.
- Wrote test cases across entire application stacks using Junit and pytest ensuring optimized test scenarios and wrote automation scripts using Selenium.
- Performed API Testing and Mobile Testing using SOAPUI, Rest Assured and Appium.
- Integrated automated tests into DevOps pipelines using Jenkins and GitHub Actions, ensuring seamless execution and reporting.

Software Engineer (Remote), Green InfoTech - Union City, CA, USA July 2025 - March 2026

- Built full-stack applications with Python and JavaScript backends and React.js, TypeScript, and CSS frontends to automate paperwork and clinical workflows for healthcare providers.
- Utilized AWS for app deployment and performance monitoring, Docker for containerization, and Terraform for provisioning infrastructure, reducing monthly costs by 10%.
- Configured CI/CD pipelines and version control using Git and GitHub Actions to automate building, testing and deployment.
- Built RAG systems for clinical note-taking agents and a text-to-SQL agent for patient databases; wrote 100+ complex SQL queries to train the models and improved accuracy by 25%.
- Developed an AI Call Agent for automated appointment scheduling and insurance verification using FastAPI. Reduced latency by 40% by designing RESTful APIs optimized for more specific responses.

PROJECTS

- *Automated Data Analysis and Visualization* - Developed a full-stack web application using Python (FastAPI, Pandas), React and Typescript that performs intelligent data analysis, and visualization on CSV files, reducing analysis time by 45%. Fine-tuned gpt-oss-20b model to provide AI insights on the analyses.
- *Question-and-Answer Agent for Research Papers* - Built a Q&A agent for Research papers using Graph RAGs (knowledge graphs), Cypher for data retrieval and Streamlit for deployment to identify links between multiple papers. Improved performance by 70% compared to traditional RAGs.
- *Adversarial Training for Deep Learning models* - Fine-tuned various SOTA deep learning models with over 20000 OCT images using Pytorch to detect retinal diseases. Performed adversarial training and achieved an average improved accuracy on adversarial examples by 50%.